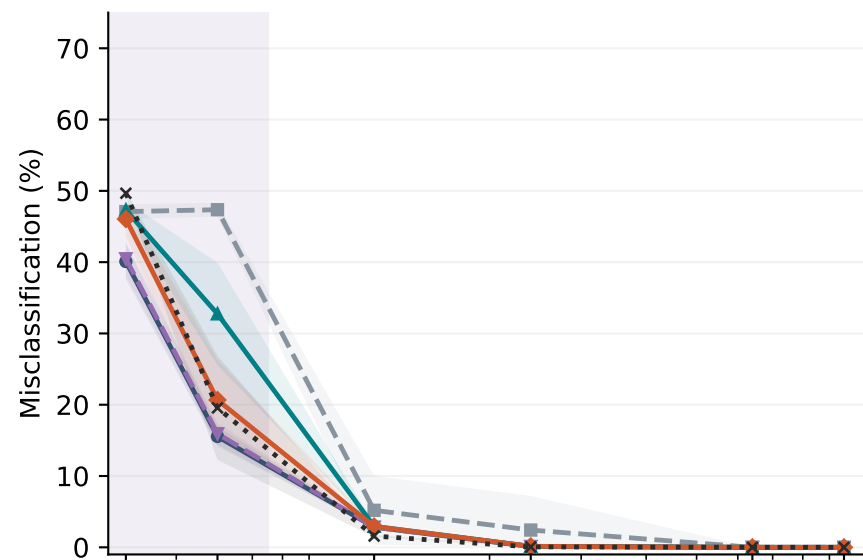


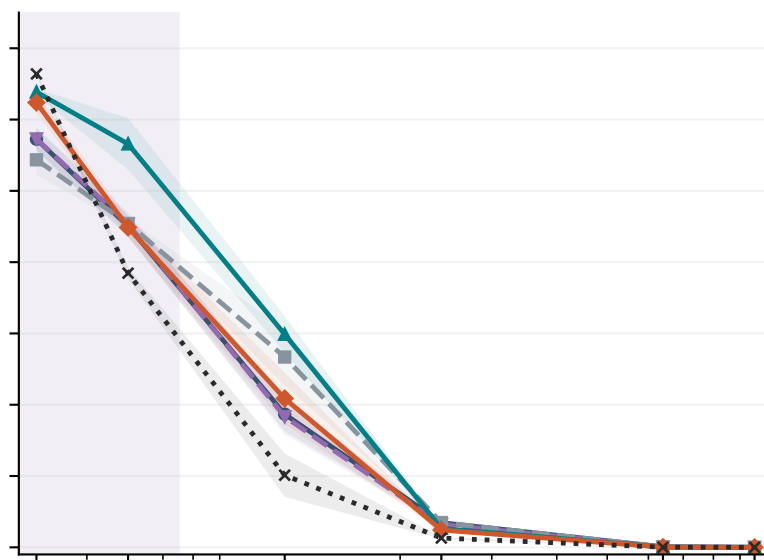
Likelihood initialization is useful, but is not uniformly better than KMeans

720 graphs · $n = 1,800$ · 20 paired graph seeds per condition · 6 starts per method

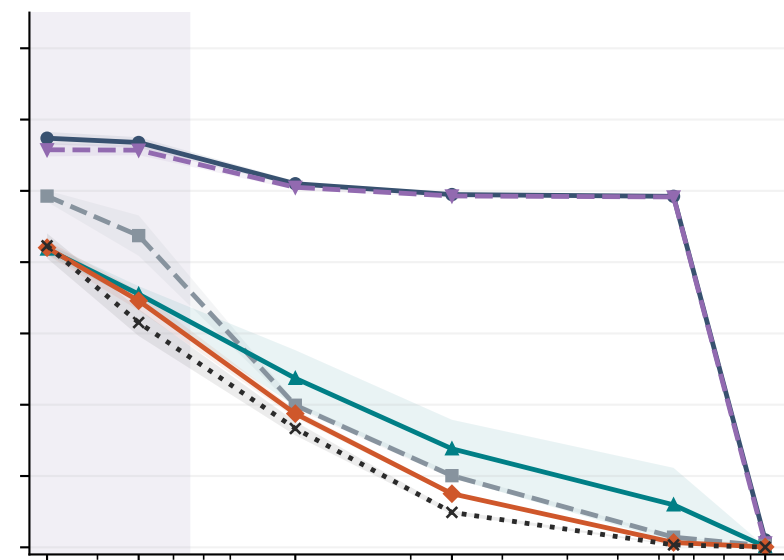
K=2: symmetric



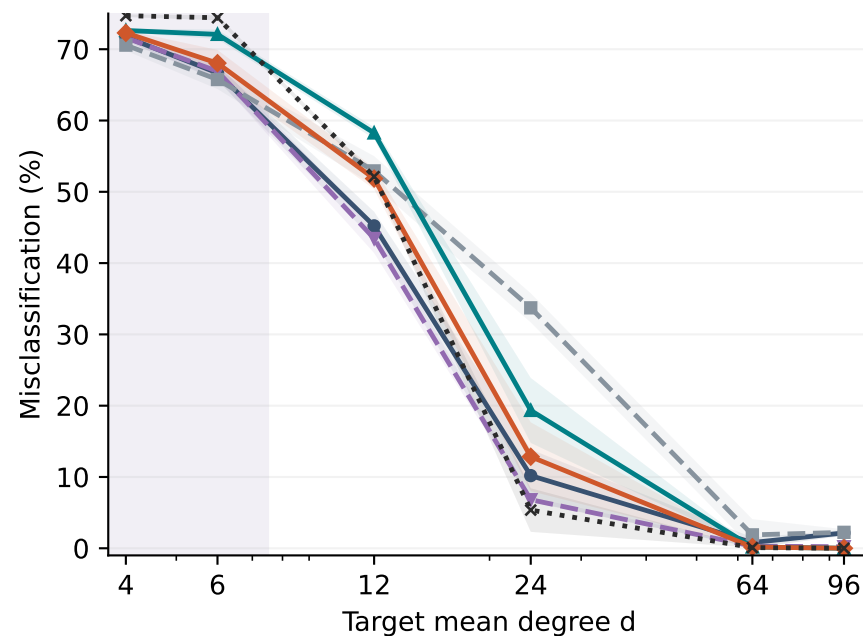
K=3: equal expected degrees



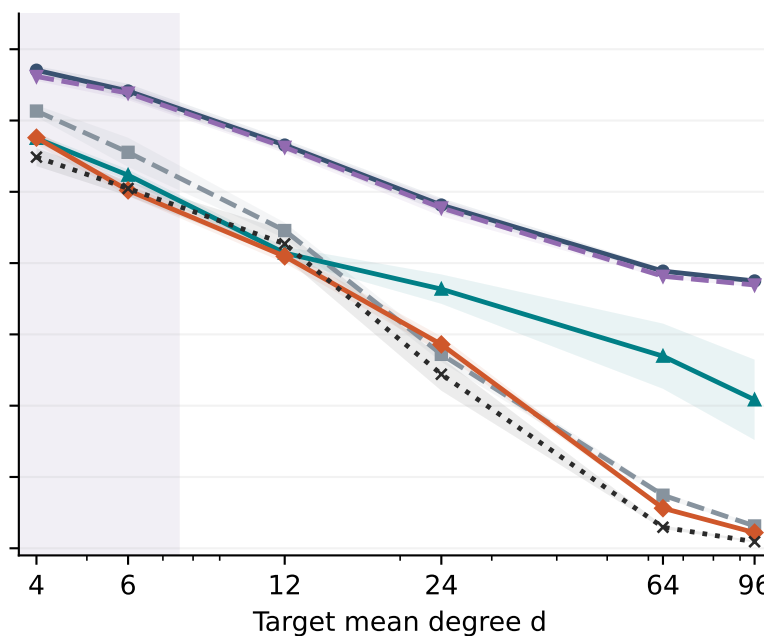
K=3: heterogeneous degrees



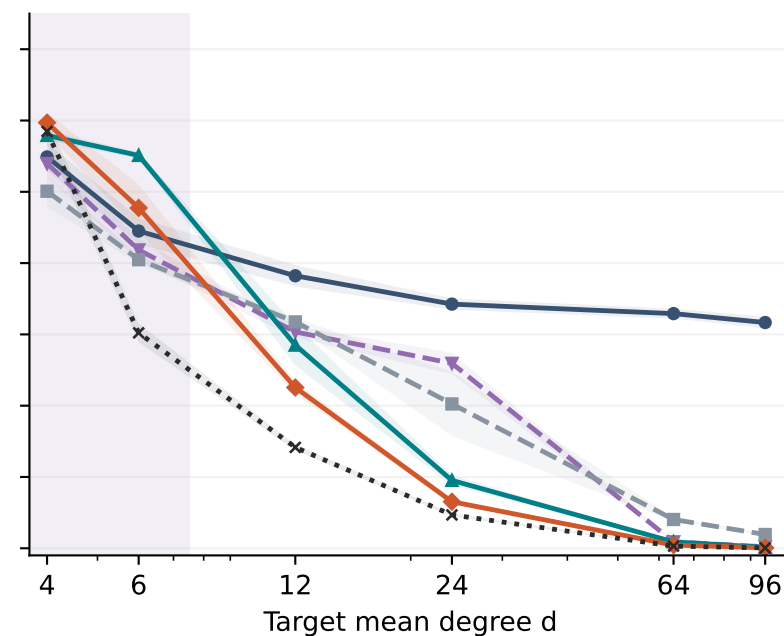
K=4: equal expected degrees



K=4: hierarchical / heterogeneous



K=3: disassortative



Shaded x region: $d < \log(n)$. Bands: pointwise 95% Monte Carlo intervals. All other curves use only degree + top-K spectrum.

—●— KMeans: eigenvectors

- -■- KMeans: neighbor mean + degree

—◆— Poisson / conditional Gaussian

- -▲- KMeans: adjacency spectral embedding

—▲— Gaussian likelihood: spectrum

- -×- Bernoulli VEM: uses raw edges